Check for updates

Received: 16 August 2023 | Revised: 15 December 2023 | Accepted: 18 December 2023

Healthcare Technology Letters

DOI: 10.1049/htl2.12073

The Institution of Engineering and Technology WILEY

LETTER

Autism spectrum disorder detection using facial images: A performance comparison of pretrained convolutional neural networks

Israr Ahmad[1] | Javed Rashid[2,3] | Muhammad Faheem[4] | Arslan Akram[3,5] | Nafees Ahmad Khan[3,5] | Riaz ul Amin[3,5]

[1]Department of Automation Science, Beihang University, Beijing, China

[2]Department of IT Services, University of Okara, Okara, Punjab, Pakistan

[3]MLC Lab, Okara, Punjab, Pakistan

[4]Department of Computing Sciences, School of Technology and Innovations, University of Vaasa, Vaasa, Finland

[5]Department of Computer Science, University of Okara, Okara, Punjab, Pakistan

Correspondence
Muhammad Faheem, Department of Computing Sciences, School of Technology and Innovations, University of Vaasa, 65200 Vaasa, Finland.
Email: muhammad.faheem@uwasa.fi

Abstract

Autism spectrum disorder (ASD) is a complex psychological syndrome characterized by persistent difficulties in social interaction, restricted behaviours, speech, and nonverbal communication. The impacts of this disorder and the severity of symptoms vary from person to person. In most cases, symptoms of ASD appear at the age of 2 to 5 and continue throughout adolescence and into adulthood. While this disorder cannot be cured completely, studies have shown that early detection of this syndrome can assist in maintaining the behavioural and psychological development of children. Experts are currently studying various machine learning methods, particularly convolutional neural networks, to expedite the screening process. Convolutional neural networks are considered promising frameworks for the diagnosis of ASD. This study employs different pre-trained convolutional neural networks such as ResNet34, ResNet50, AlexNet, MobileNetV2, VGG16, and VGG19 to diagnose ASD and compared their performance. Transfer learning was applied to every model included in the study to achieve higher results than the initial models. The proposed ResNet50 model achieved the highest accuracy, 92%, compared to other transfer learning models. The proposed method also outperformed the state-of-the-art models in terms of accuracy and computational cost.

1 | INTRODUCTION

The ability to process and understand information is significantly impacted by autism spectrum disorder (ASD), a complex neurological and developmental condition. The incidence of ASD among individuals of all ages is currently increasing at an alarming rate nowadays. Typically, children with this disorder begin displaying symptoms between the ages of 2 and 5 [1]. Initially observed in young children, the phenomenon has since spread to adolescents and adults. Autism is a severe neuro developmental disorder characterized by difficulties in social interaction and communication and the presence of restricted and repetitive behaviours [2]. A decline in verbal competence is often observed in individuals with ASD [3]. Both neurological and genetic factors are linked to the development of this disorder. Social interaction, reasoning, visualization, repetitive behaviours, and problems in social interaction are all indicators of ASD [4]. At present, there is no cure for this syndrome. The exact causes of ASD are not yet fully understood, but it is believed that both environmental factors and genetic abnormalities may play a significant role in the progression of this disease. Several genetic mutations [5] have been linked to ASD, affecting multiple brain regions. The introduction should be succinct, with no subheadings. Limited figures may be included only if they are truly introductory and contain no new results.

It is estimated that 1 in 44 American children has been diagnosed with autism spectrum disorder (ASD) based on data from the Centres for Disease Control and Prevention (CDC) [6]. The incidence of autism has risen at an alarming rate in recent years, with estimates suggesting that it affects one in every

This is an open access article under the terms of the Creative Commons Attribution License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.
© 2024 The Authors. *Healthcare Technology Letters* published by John Wiley & Sons Ltd on behalf of The Institution of Engineering and Technology.

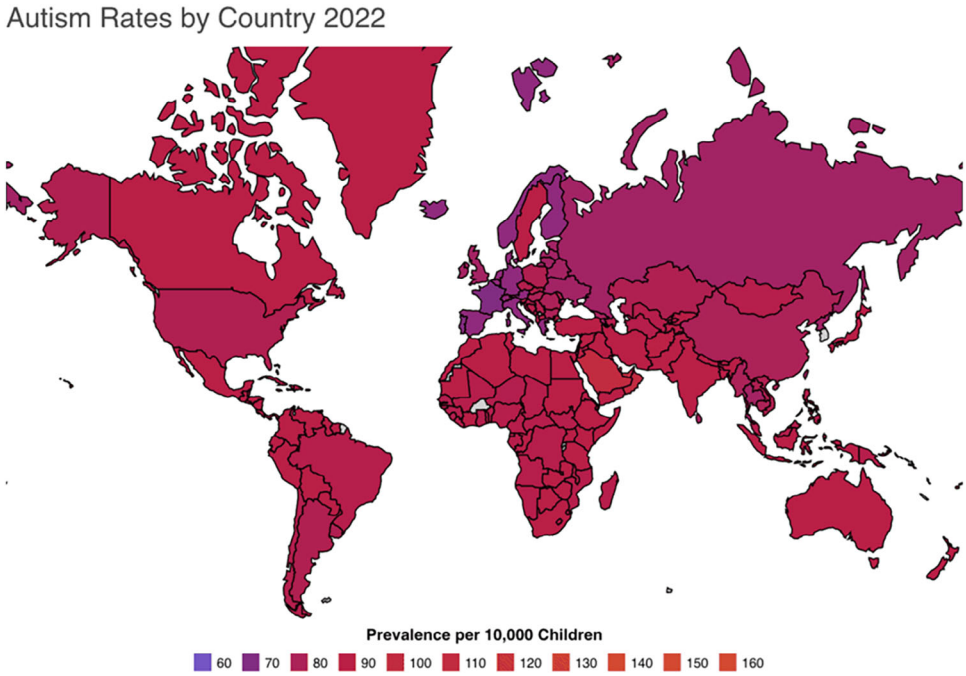

FIGURE 1 ASD spread rates country wise 2022 [12].

70 children born today [7]. Boys are more likely to be diagnosed with ASD than girls are [6]. During the study period 2009–2017, parents of about 1 in 6 (3–17-year-old) children reported that their child was diagnosed with a developmental disability. Disorders including autism, ADHD, blindness, and cerebral palsy were listed [8]. It has been noted that ASD affects people of every race, ethnicity, and socioeconomic status [6]. Research has shown that multiple gene mutations account for 10–20% of ASD patients [9, 10], and more than a hundred genes have been connected to the disorder. The patient's mental and physical well-being can be better preserved with an early diagnosis of the neurological disease [11]. A study conducted by CDC suggests that in 2022, the rate of ASD increased by 178% in the US only. The US states are reporting a higher rate of ASD in history; according to a close estimation, 2.2% of the whole population is autistic. 75% of autistic adults are unemployed, and autism leads to other mental health conditions in 78% of children [12]. The estimated caring cost of US persons diagnosed with only autism disorder was 268 billion US dollars in 2015. The care cost for autistic persons will be 461 billion US dollars which is almost double the amount in 2015 [12] (see Figure 1).

Furthermore, children with other mental illnesses, in addition to autism, will require even more expenditure for their care. These statistics were only from the United States, which is a developed country with excellent healthcare facilities; other developing nations will face even greater problems. The global prevalence of autism disorders is increasing; as reported by the CDC, 2.84% of the population of South Korea is autistic, 1.14% of the population of Qatar is autistic, and in California, the ASD rate is 3.9% [13]. In light of these statistics, due to the increasing prevalence of ASD, other mental health conditions are also on the rise, making the lives of individuals with autism even more severe, and requiring additional funds and resources for their care. Early diagnosis of ASD in children can reduce its effects on individuals and ultimately help countries reserve more funds for research on other fatal diseases instead of spending on the care of individuals with autism [13].

A lack of eye contact, stereotypical behaviour, and poor social interaction are key characteristics of this order, which is often identified in children over the age of 2 [14]. However, there are currently no detectable clinical tests for this syndrome, such as a blood test, making the diagnosis a challenging task. To address this issue, several psychological questionnaires have been developed to fill the void left by the absence of pathophysiological diagnostic criteria indicative of ASD. These questionnaires include age appropriate questions to assess the child's social interaction and behaviour level. The diagnosis is confirmed with the aid of these instruments, as well as the patient's medical history, clinical observations, and IQ tests [15]. An early diagnosis is essential for providing timely therapies to alleviate the symptoms of ASD and to enable the child to develop the abilities necessary to function later in life, even though the disorder is still incurable.

With the advancement of machine learning (ML) and deep learning (DL) techniques, these have been widely applied in various fields, such as lyme rashes disease recognition [16, 17], facial expression identification [18, 19], sentimental analysis [20], health monitoring [21, 22], gender classification [23] and medical disease diagnosis [24–26]. Numerous studies [27, 28] analysed the significance of different characteristics that might be used to diagnose ASD, such as behavioural patterns, facial appearance, eye tracking, and speech. However, these

20533713, 2024, 4, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/htr2.12073 by PEC University Of Technology, Wiley Online Library on [23/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

characteristics can be subtle and elusive when observing a young child. Additionally, healthcare practitioners may not possess the necessary knowledge or resources to make a diagnosis or have difficulty communicating with autistic youngsters. The promising results of disease prediction utilizing deep learning-based algorithms surpass human evaluation of large datasets. Furthermore, very little in the way of automated or deep learning-based models for diagnosing ASD from observed behaviours and physical characteristics currently exist. Using facial photos of autistic and non-autistic children from the Kaggle dataset of autism disease [28], various pre-trained convolutional neural network architectures were utilized to gain insights about this condition.

2 | RELATED WORK

Over the past few years, a growing body of research has focused on analyzing facial expressions in people with ASD. Patients with neurological illnesses, such as frontotemporal dementia [29], neurodegenerative disorders [30], and Alzheimer's [28], have been studied extensively concerning facial expressions. To help children with ASD understand and use a range of facial expressions, many approaches have been offered. With the help of a CNN, Yolcu et al. [31] were able to determine which face characteristics were significantly associated with ASD. The appropriate facial expressions were identified using a second CNN model. This model reached the best prediction rate of 94.44%. The same team of researchers subsequently refined the model by training it to recognize facial features such as lips, eyes, and eyebrows in kids with neurological problems. Differential diagnosis was made more accessible by integrating these models into iconized visuals.

Six universal emotions were compared between children with and without ASD using time-series modelling and statistical analysis by Guha et al. [32]. Children with ASD were found to have fewer nuanced facial expressions, especially around the eyes. Using a 3D virtual environment, Grossard et al. [33] created an educational multimodal emotional imitation game (JEMImE) to help children with ASD express emotions like grief, rage, and happiness. The game's various visual and motivational features helped draw the kids in [34].

To evaluate human facial expressions, Valles et al. [35] trained CNN models on photographs from Kaggle's (FER2013) 2013 dataset, which had been adjusted to include images of children with ASD taken in a variety of lighting conditions. Emotional assessments of children with ASD receiving robot-assisted treatment (RAT) were also the subject of several studies [36, 37], which used deep learning and Raspberry Pi3 models.

The effectiveness of a portable motion detector for children with ASD was evaluated by Smitha et al. [38], who contrasted a parallel and serial principal component analysis (PCA) feature extraction approach. The PCA method [39] was implemented in hardware to ease the extraction of helpful motion features, and it reached a maximum accuracy of 82.3% for a word length of 8 bits. To better recognise image-based facial expressions associated with ASD, Pramerdorfer et al. [40] compared the effectiveness of different CNN models. The software's detection performance was much enhanced by the incorporation of fundamental CNN architectures, as compared to that of competing models. However, the accuracy of a recent deep CNN ensemble was 75.2%, which was far superior to that of any previously developed models. In addition, this model does not need additional data for face registration or training, which is a huge time saver.

Ibala et al. [41] employed transfer learning and leveraging ensemble approaches to classify face photos of people with and without ASD according to seven core human emotions: anger, neutrality, fear, happiness, surprise, disgust, and sadness. Maximum accuracy was reached at 78.3% using transfer learning and 67.2% using ensemble approaches. A prototype deep learning system for automated ASD diagnosis was developed by Niu et al. [42]. With regards to ASD classification, their model was 73.2% accurate. Using a combination of personal distinguishing data and functional connectomes from all three tiers of the brain, they were able to create a model that outperformed many other machine learning models. There was promise in these findings for developing deep learning models. Their study demonstrates the potential for deep learning frameworks to aid in the development of automated clinical diagnostics for ASD in the future.

The ASD screening model proposed by Thabtah [43] used machine learning techniques. The advantages and disadvantages of several classification methods for individuals with ASD were examined in their study. Problems with ASD screening methods were brought to light, such as using the DSM-IV manual instead of the DSM-5 text. ASD categorization was investigated by Mythili and Shanavas et al. [44]. The primary goal of their study was to identify autistic symptoms and quantify their severity. In order to examine the kids' social behaviour and interactions, they have employed the data mining tool WEKA and the support vector machines (SVM) and fuzzy logic techniques. Li et al. [45] examined adults with autism identified by an imitation-based machine-learning classification strategy. Their primary goal was to examine the issues at the heart of discriminatory testing conditions and kinematic factors. Their findings suggest that machine learning techniques could be helpful for autism screening and diagnosis.

Numerous deep learning methods have been developed to address the wide range of categorization issues present in medical images. In this research, we have attempted to utilize face images for the diagnosis of ASD by employing several pre-trained convolutional neural network architectures.

3 | MATERIALS AND METHODS

This study proposed an autism spectrum disorder detection using facial images (ASD-DFI) at early age. The methodology contained 6 greatly acknowledged pre-trained models. ASD-DFI was trained and tested on ASD image data images dataset.

20533713, 2024, 4, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/hit2.12073 by PEC University Of Technology, Wiley Online Library on [23/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

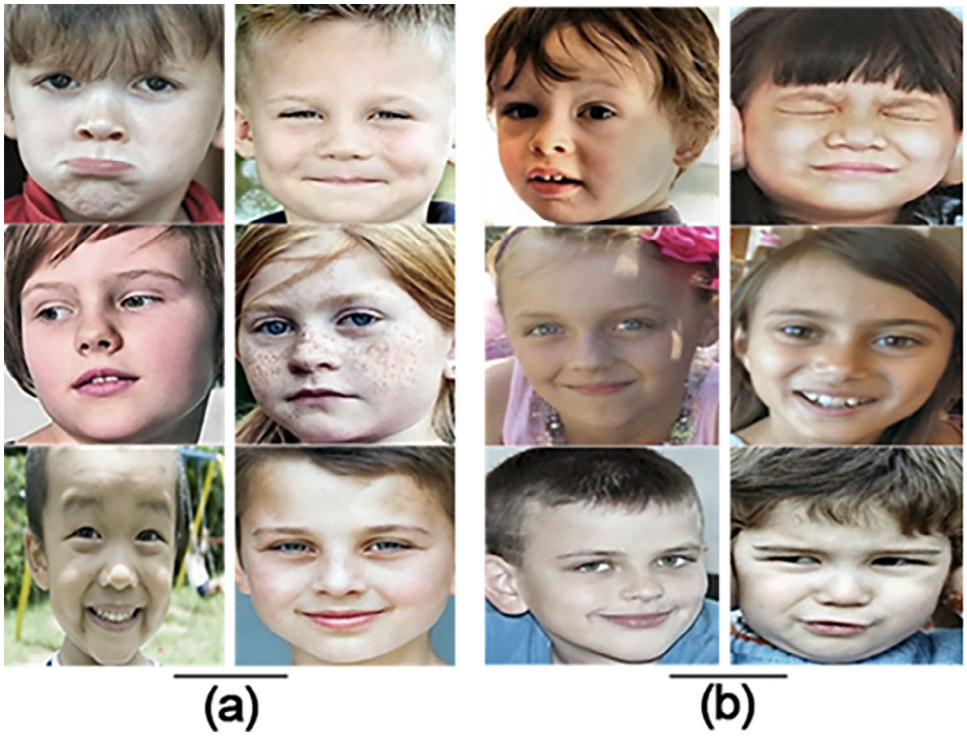

FIGURE 2 (a) Positive ASD samples, (b) negative ASD samples.

3.1 | Autism spectrum disorder dataset

In the current study, a publicly accessible dataset for ASD was used [46]. The dataset can be used in two ways; one is in training, validation, and testing form and other is in consolidation form where a random split required in data loader before feeding to CNN. The consolidation section contains two subdirectories of autistic and non-autistic. Similarly, other form also contains sub two sub directories of autistic and non-autistic for training, validation, and testing. Sample images of dataset are presented in Figure 2.

Where each category comprises correspondent images with the same dimensions and each image was loaded as 3 channels image. Almost every image in the dataset is captured while posing straight face in front of camera. The dataset contained a total of 2940 images for both autistic and non-autistic, where the number of each class were equal, it means the dataset was balance. For this study, the dataset was saved in to two different settings, in first setting dataset was saved in 124 × 124 pixels for each, and for second 248 × 248 pixels for each image to imitate grid search for investigation of the relation between performance and pixels sizes of the images. Their details are described in Table 1.

TABLE 1 Summary of ASD dataset.

Mode	Dimensions	Classes	Images
Setting 1	124 × 124	Autistic	1470
		Non-autistic	1470
Setting 2	248 × 248	Autistic	1470
		Non-autistic	1470
Total			**5880**

3.2 | Data preprocessing

Data processing is an important step in preparing data for use in machine learning and other forms of analysis. It helps to ensure that the data is in a consistent format and that any outliers or errors are removed. This can lead to better results when using the data for classification or other types of analysis. In the case of the ASD dataset, image processing was applied to the data in order to improve the results of the analysis. Three specific preprocessing techniques were used on the ASD dataset.

The first technique was to resize all of the images in the dataset. There are several ways in which resizing images in deep learning models improves the model's performance. First, it improves computational efficiency by making training and inference less taxing on the computer, which speeds up processing and uses less memory. Because of this, the model can converge faster during training as a whole. When photos are resized, the model learns robust features across different image sizes, improving its generalization and ability to recognize things in different situations. For models to be deployed in production, resizing inputs to standardize their sizes is essential. It will provide consistency and facilitate integration across many applications. By utilizing effective data augmentation techniques,

20533713, 2024, 4, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/hit2.12073 by PEC University Of Technology, Wiley Online Library on [23/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

smaller image dimensions save memory needs and make models more adaptive to real-world settings. Reducing image sizes leads to lower storage requirements, making this strategy especially significant in applications with limited storage space. Reducing the size of images has several benefits, including making better use of computational resources, training more quickly, increasing generalizability, and making deployment more flexible. This was done in two different settings (124 × 124 and 248 × 248) in order to compare the performance of pre-trained models. Python code was used to resize the images in two different settings.

The second technique was to flip some of the images in the dataset left and right in order to add more diversity to the data. For deep learning models, image flipping has clear benefits, especially when it comes to horizontal and vertical flips. Firstly, image flipping is a highly effective data augmentation strategy for model training. Incorporating different object orientations improves the model's generalizability to different situations. It is quite helpful for computer vision applications like object identification and image classification, where the model needs to identify things from various angles. Flipping also helps the model be more resilient by decreasing overfitting and exposing it to diverse training instances. To further improve the model's ability to identify items in any orientation, flipping also helps to incorporate rotational invariance. This method shines when working with sparsely labelled data since it gives the model a better grasp of item variability by artificially expanding the dataset. To conclude, picture flipping is an effective training pipeline tool for deep learning models since it generally improves generalization, robustness, and performance. For flipping the images data generator utility of Keras is applied.

The third technique was to apply a normalization step to all of the images in the dataset. Specifically, the ImageNet statistics ([0.485, 0.456, 0.406], [0.229, 0.224, 0.225]) were used to normalize the images. One of the most important preprocessing steps in building deep learning models image normalization has many benefits. Image normalization primarily involves adjusting pixel values to a predetermined range, typically from 0 to 1. This standardization ensures that model training is constant and stable by removing the impact of images with varied intensities. Normalization speeds up convergence while training because it prevents problems like exploding or vanishing gradients that occur when pixels have very wide value ranges. In addition, preserving numerical stability improves the model's capacity to learn significant characteristics. The procedure helps to reduce the effect of extreme values and changes in light, making the model more adaptable to varying lighting scenarios. Because pre-trained models are typically developed on datasets with standardized input, normalization also makes it easier to use these models effectively. Deep learning models can perform better across various applications because of picture normalization, improving their efficiency, stability, and generalization capabilities.

These preprocessing steps were applied to the dataset in order to improve the results of the analysis and to ensure that the data was in the best possible format for use in machine learning and other forms of analysis.

TABLE 2 ASD dataset splitting summary.

Mode	Classes	Training	Validation	Testing	Total
Setting 1	Autistic	1176	147	147	1470
	Non-autistic	1176	147	147	1470
Setting 2	Autistic	1176	147	147	1470
	Non-autistic	1176	147	147	1470
Total		**4704**	**588**	**588**	**5880**

3.3 | ASD dataset splitting

In this study the first way of ASD dataset was used where it was physically divided into two parts, training, and testing. Regardless of the original distribution, the dataset was divided into a ratio of 80:10:10, with 80% of the data allocated for training, 10% for validation, and 10% for testing. For the autistic class, 1176 images were used for training, while an equal number of images were used for the non-autistic class. For validation, 147 images were used for the autistic class and the same number for the non-autistic class. For testing, both the autistic and non-autistic classes had 147 images per class. The total images for training were 2940, where 2352 were for training, 294 for validation and 294 for testing. The detailed summary for splitting is presented in Table 2.

3.4 | Proposed methodology

Convolutional neural networks (CNNs) have established a strong reputation in the field of image data processing, producing superior results compared to traditional methods. However, they require enormous data for the training phase, sometimes socially referred to as data-hungry algorithms especially training from scratch. However, transfer learning (TL) has solved this problem to a very large extent, where a pre-trained model is retrained to perform a specific task with fewer data samples. Pre-trained deep learning models provide substantial benefits in artificial intelligence and machine learning. These models save practitioners time and computing resources by providing powerful starting points for numerous tasks, leveraging knowledge from prolonged training on big and diverse datasets. The ability to adapt pre-trained models to specific tasks using limited labelled data is a major feature of transfer learning. It reduces the requirement for enormous datasets. Pre-trained models are flexible because they can parse many kinds of data for useful hierarchical characteristics. Additionally, these models are useful for practitioners in several areas due to their resistance to overfitting and rapid convergence during fine-tuning. Efficient deployment of complex architectures in various applications, from computer vision to natural language processing, is made possible by the availability of pre-trained models from well-established repositories, encouraging community cooperation. In general, pre-trained deep learning models have the potential to democratize access to cutting-edge machine learning technology, be more efficient, and have high transferability.

20533713, 2024, 4, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/htl2.12073 by PEC University Of Technology, Wiley Online Library on [23/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

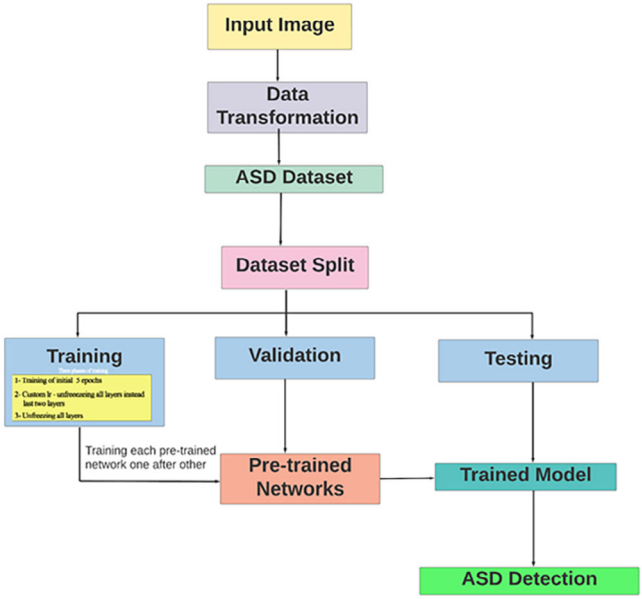

FIGURE 3 Flowchart of the proposed method.

The selection of the best TL model is crucial, as different models may perform better or worse depending on the dataset. One approach to address this enigma is employing top networks as a whole to achieve good results. In this research, we employed six widely recognized pre-trained models, ResNet34, ResNet50, VGG16, VGG19, AlexNet, and MobileNetv2 - to evaluate their performance in the detection of autistic disorders based on facial images. To this end, we employed the ResNets models with two different settings of the ASD dataset, utilizing image sizes of 124 × 124 and 248 × 248 pixels to investigate the relationship between performance and image size. The training of all these pre-trained models was completed in three batches of epochs, with each model undergoing a total of 5 initial epochs, followed by a selective learning rate for an additional seven epochs, during which all layers were frozen except for the last two. Finally, each model was unfrozen for further training for a total of 8 epochs. This approach allows the models to learn from the data more thoroughly and efficiently and allows for the fine-tuning of the models using the selective learning rate. The block architecture of the proposed methodology is illustrated in Figure 3.

Image analysis can classify autistic people by identifying distinguishing visual traits. Here, facial expressions are important since they reveal emotions and social interactions. Eye gaze patterns are important because autistic people may have unusual eye contact. Body language, including gestures and postural clues, helps explain autism's peculiar behaviours. A holistic view of an individual's relationships comes from contextual awareness, which includes social and environmental aspects. Recognition of repetitive actions, a prevalent autistic feature, and social interaction dynamics aid classification. Machine learning and deep learning can extract and evaluate these properties from photos, revealing autism's complex spectrum. This research must be sensitive to autism heterogeneity and follow ethical norms for visual data classification [14].

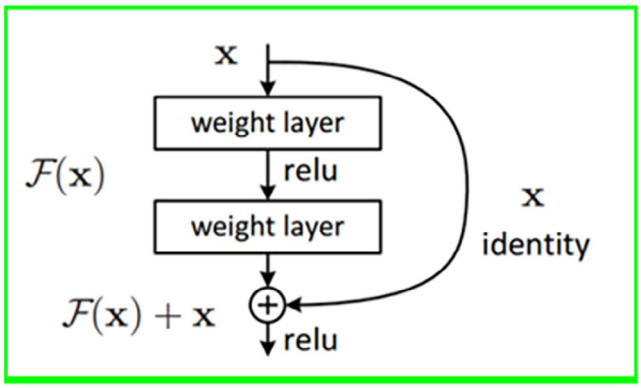

FIGURE 4 Residual network block.

The goal of this research is to train and identify the best model based on the given images and subsequently propose the most effective model for the early detection of autism spectrum disorder (ASD). A description of the pre-trained networks is provided below.

ResNet Network: ResNet, or the residual network, is a pioneering convolutional neural network (CNN) that utilizes the concept of "residual blocks" to mitigate the challenge of vanishing gradients in training extremely deep networks, as depicts in Figure 4. The idea behind ResNet is that instead of letting layers learn the underlying mapping, let the network fit the residual mapping. The residual block comprises multiple layers, where the output of these layers is added to the input before it is passed on to the subsequent block. This allows the network to learn residual representations of the input data instead of trying to fit the input data directly. The approach is to add a shortcut or a skip connection that facilitates the flow of information from one layer to the next. ResNet's residual block is multilayered, mixing its output with the original input before passing on. In numbers:

$$X_{\text{out}} = F(X_{\text{in}}) + X_{\text{in}} \tag{1}$$

where, X_{in} is the input to the residual block. F_{in} represents the transformation applied to the input by the layers within the block. X_{out} is the output of the residual block, which is the sum of the transformed input and the original input? By adding the input to the changed output, ResNet efficiently learns the residual mapping, allowing the network to focus on learning the "difficult" parts of the mapping rather than the complete mapping from scratch. This residual learning makes training incredibly deep networks with improved performance significantly easier.

AlexNet: AlexNet is a convolutional neural network (CNN) model that was introduced in 2012 by Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton. It was the first deep learning model to be trained on a massive dataset, the ImageNet, which contains more than a million images from 1000 different classes.

20533713, 2024, 4, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/hit2.12073 by PEC University Of Technology, Wiley Online Library on [23/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

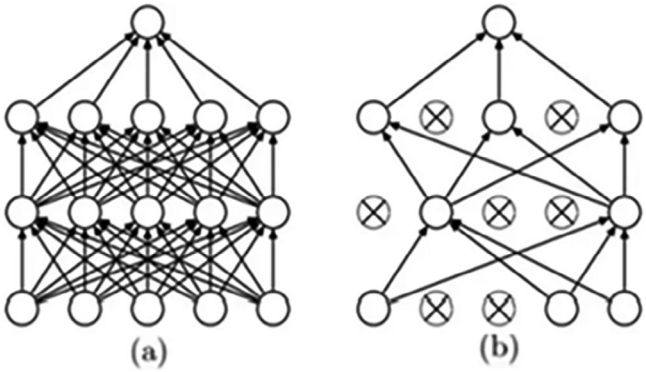

FIGURE 5 AlexNet architecture.

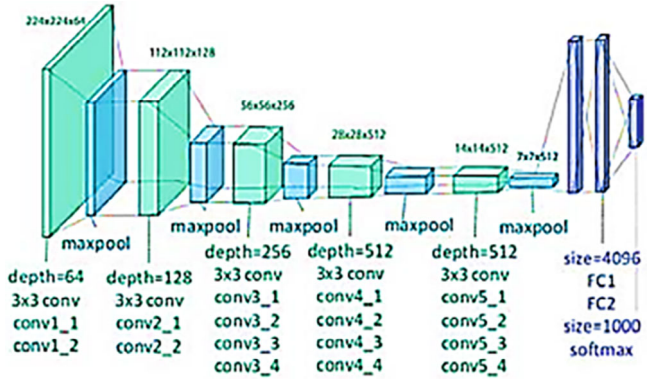

FIGURE 6 VGG19 architecture.

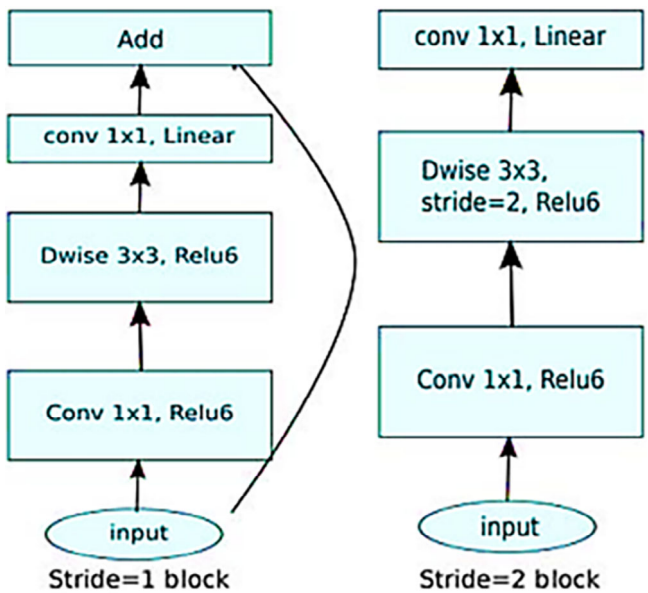

FIGURE 7 MobileNetV2 architecture.

AlexNet achieved a top-5 error rate of 15.3%, which was a significant improvement over the previous state-of-the-art models. The architecture of AlexNet consists of eight layers, including five convolutional layers and three fully connected layers, as shown in Figure 5.

Let's discuss the math behind AlexNet's framework: the equation for convolutional layer is as follows:

$$Z^l = W^l * A^{l-1} + b^l \tag{2}$$

$$A^l = \mathrm{ReLU}\left(Z^l\right) \tag{3}$$

where Z^l is the output of the convolutional layer, W^l is the filter weights, A^{l-1} is the input feature map from the previous layer, and b^l is the bias term.

The equation for max pooling is:

$$A^l = \mathrm{MaxPooling}\left(A^{l-1},\ \text{filter size},\ \text{stride}\right)$$

where, MaxPooling is the max pooling operation.

For the fully connected layer:

$$Z^l = W^l . A^{l-1} + b^l \tag{4}$$

$$A^l = \mathrm{ReLU}\left(Z^l\right) \tag{5}$$

$$A^l_{\mathrm{Dropout}} = \mathrm{Dropout}\left(A^l,\ \text{dropout rate}\right) \tag{6}$$

where Z^l is the output of the convolutional layer, W^l is the filter weights, A^{l-1} is the input feature map from the previous layer, and b^l is the bias term, and dropout is dropout operation.

Softmax function for the output layer is as:

$$P\left(y = i|x\right) = \frac{e^{z_i}}{\sum_{j=1}^{1000} e^{z_j}} \tag{7}$$

VGG19: VGG (visual geometry group) network is a deep convolutional neural network model that was introduced in 2014 by researchers at the University of Oxford. The VGG network architecture is known for its use of very small convolutional filters (3 × 3) and stacked multiple layers of convolutional and max-pooling layers. This architecture was designed to increase the depth of the network, which allows the model to learn more complex features from the images, as depicts in Figure 6.

MobileNetV2: MobileNetV2 is a lightweight convolutional neural network (CNN) architecture designed for mobile and embedded devices with limited computational resources. It was introduced in 2018 by Google researchers Andrew Howard, Mark Sandler, and Grace Chu. MobileNetV2 improves upon the original MobileNet model by utilizing depthwise separable convolutions, which reduces the number of parameters and computation required, making it more efficient for mobile devices. It also introduced the concept of linear bottlenecks, which further reduced computation while maintaining accuracy. The architecture of the MobileNetV2 shown in Figure 7. Here is a mathematical expression of the skip connection and the depthwise separable convolutions:

$$Y_{i,j,k} = \sum_{m,n} X_{i+m,j+n,k} \times K_{m,n,k} \tag{8}$$

20533713, 2024, 4, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ntt2.12073 by PEC University Of Technology, Wiley Online Library on [23/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 3 Pre-trained networks, number of convolutions layers summary.

Deep Architecture	Layers	Trainable parameters
ResNet34	34	21 M
ResNet50	50	25.5 M
AlexNet	8	61 M
InceptionNetV2	53	3.4 M
VGG16	16	138 M
VGG19	19	144 M

X represents the input feature map and Y represents the intermediate feature map after the depthwise convolution

In this study, we employed a transfer learning approach for the detection of autism spectrum disorder (ASD) utilizing pre-trained networks that have been previously trained on the ImageNet dataset. This approach aims to leverage the pre-existing knowledge of low-level feature extraction, such as edges, boundaries, and shapes, to extract high-level features based on shared knowledge. This process, known as transfer learning in deep learning, enables the efficient utilization of pre-existing knowledge to improve performance on a new task. Each network employed in this study has a specific number of layers and parameters. The number of layers and the number of learnable parameters for each model, which provides insight into the number of filters used by the models, is presented in Table 3. This information can provide valuable context for understanding each model's relative complexity and capacity and can inform the selection of the most appropriate model for a given task.

3.5 | Evaluation measures

The metrics for measuring performance used in this experiment are accuracy, precision, recall, and error rate. One epoch represents a complete pass of all available training sets, which are iterated in batches. The evaluation measures for the experiments are the following:

The accuracy of a classifier is calculated as a ratio of positive to the total number of positive samples (either correctly or incorrectly).

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{9}$$

Images that have been successfully identified as tampered with are true positives (TP), while those that have been incorrectly identified as such are false negatives (FN). Images that have been accurately classified as true negatives (TN) and incorrectly classified as false positives (FP) are shown. Misclassification occurs when either a manipulated image or a legitimate image is incorrectly labelled as genuine.

Error: The model that makes all the incorrect predictions is called error. This is used to know about all the wrong predictions.

$$\text{Error} = \frac{\text{FP} + \text{FN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}} \tag{10}$$

Precision: Precision (also called positive predictive value) is the fraction of relevant instances among the retrieved instances. It is a measure of how the model works in all classes. It is useful if all classes are equally important. The ratio of the right estimates to total projections is determined. The precision is defined as:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \times 100 \tag{11}$$

Recall: Recall (also known as sensitivity) is the fraction of relevant instances that were retrieved. It is also a critical parameter for the evaluation of the performance of a model. The proportion of input samples from a class that the model correctly predicts. Calculating the recall is as simple as:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \times 100 \tag{12}$$

F1-Score: When evaluating a model, the F1 score considers both the recall and precision metrics. The minimum is zero, and the highest is one. If the value is the maximum, then the model is optimal.

$$\text{F1} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \times 100 \tag{13}$$

4 | RESULTS AND DISCUSSION

The proposed method for detecting autism spectrum disorder (ASD) was evaluated through a series of experiments. The study, 90% of the data was utilized for training and validation, while the remaining 10% was reserved for testing. To assess the performance of all models in the detection of ASD, each model was trained and subsequently tested. The experimental results focused on the following aspects:

1. Comparison of results in terms of accuracy for all models, presented in tabular form.
2. Detailed discussion of the results of the proposed ASD-DFI method.
3. Comparison of the performance of the ASD-DFI method in terms of accuracy to state-of-the-art models, in order to highlight the significance of the proposed method.

The experiments were executed using a Google Collaboratory [47] pro account with an Nvidia T4 GPU [48] and 16 gigabits of additional RAM. The study utilized version 5.3 of FastAiV2 [49] with PyTorch [50] as the backend, and the batch size was set to 64. The Adaptive Moment Estimation (Adam) optimizer was employed to handle sparse gradients on noisy snags, while the Flattened Loss function was used to calculate the losses. The training time (runtime) was approximately 2 h, and the testing time was 2 to 3 min.

20533713, 2024, 4, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/hit2.12073 by PEC University Of Technology, Wiley Online Library on [23/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 4 Pre-trained networks accuracy and error rate results.

Model	Input size	Setting 1		Setting 2		Setting 3	
		Accuracy (%)	Error (%)	Accuracy (%)	Error (%)	Accuracy (%)	Error (%)
ResNet34	124 × 124	76	24	83	17	84	16
ResNet50		79	21	82	18	87	13
VGG16		79	21	81	19	81	19
VGG19		81	19	85	15	87	13
MobileNet		79	21	82	18	85	15
AlexNet		77	23	75	25	75	25
ResNet34	248 × 248	81	19	83	17	87	13
ResNet50		80	20	85	15	92	8

4.1 | The results comparison of pre-trained networks

In the initial stage of training, it was observed that all of the network models achieved an accuracy score of greater than 75%. However, the ResNet family displayed good performance, with an accuracy score of 80% in their initial training batches. Specifically, the ResNet34 and ResNet50 models exhibited the lowest error rate among all of the other networks. When utilizing a resolution of 124 × 124, both the ResNet34 and ResNet50 models achieved maximum accuracy scores of 76% and 79%, respectively, during the initial 5 epochs of training. The VGG16 and VGG19 models also performed well in comparison to the ResNet models, attaining accuracy scores of 78% and 80%, respectively. Despite the small number of layers, there was a significant difference in the number of parameters between these models. The MobileNetV2 and AlexNet models achieved accuracy scores of 78% and 76%, respectively. Furthermore, when utilizing a resolution of 248 × 248, the ResNet34 and ResNet50 models both achieved an accuracy score of 80%.

After the first 5 epochs, a selected learning rate was applied for 7 additional epochs, during which all layers were unfrozen except for the last two layers. This approach allowed the models to begin training from the start, without utilizing the low-level knowledge previously extracted from the ImageNet dataset. However, the last 2 layers were not trainable, instead they utilized the high-level knowledge acquired during the first training batch. The learning rate for each model was determined using the intrinsic FastAi function known as find lr.

On the custom learning rate, the ResNet50 model with 248 × 248 dimensions outperformed all other models, achieving an accuracy score of 85%. The ResNet34 model on 248 × 248 attained an accuracy score of 82%. On a resolution of 124 × 124, the ResNet34 and ResNet50 models produced accuracy scores of 83% and 82%, respectively. The VGG16 and VGG19 models achieved accuracy scores of 80% and 84%, respectively. The AlexNet and MobileNetv2 models produced accuracy scores of 74% and 82%, respectively. The detailed results for each model are presented in Table 4.

Upon completion of the second batch of training epochs, all layers of the models were unfrozen and made available for training for an additional 8 epochs. The ResNet50 model with a resolution of 248 × 248 achieved an accuracy score surpassing 92%, thereby outperforming all other models. The second-best performing model was the ResNet34, which attained an accuracy score of 86% on a resolution of 248 × 248. The VGG16 and VGG19 models, however, could only achieve accuracy scores of 80% and 86%, respectively. The AlexNet and MobileNetV2 models maintained their previous performance, attaining accuracy scores of 74% and 84%, respectively. When utilizing a resolution of 124 × 124, the ResNet34 and ResNet50 models achieved accuracy scores of 83% and 87%, respectively. In light of these results, it is clear that the ResNet50 model surpasses all other models. As such, we propose this model for the detection of autism spectrum disorder (ASD) using images in early childhood.

4.2 | Performance proposed ResNet50 for autism spectrum disorder detection from facial images (ASD-DFI)

In this study, we propose the ResNet50 model for detecting autism spectrum disorder (ASD) in early childhood. As previously outlined, the training process was carried out in three sections, incorporating all models included in the study and our proposed ASD-DFI model. Upon completion of the training, the performance of the proposed ASD-DFI model was evaluated using validation and testing data. Figure 8 presents the accuracies of the ASD-DFI model.

By interpreting the graph, we can perceive that the training accuracy began at just above 75% and progressively increased, reaching 78% after the third epoch (Figure 8). However, a slight decline was observed in the fifth epoch, marking the end of the first training batch. The validation accuracy began at 70% and reached 75% after the fifth epoch. Following this, the layers were unfrozen, with the exception of the last layers, which were kept frozen in order to retain the knowledge of the previous batch for high-level feature extraction. A custom learning rate was applied to all trainable layers, resulting in an immediate improvement in training accuracy, which exceeded 80% after the seventh epoch. The validation accuracy remained unchanged

20533713, 2024, 4, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/hit2.12073 by PEC University Of Technology, Wiley Online Library on [23/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

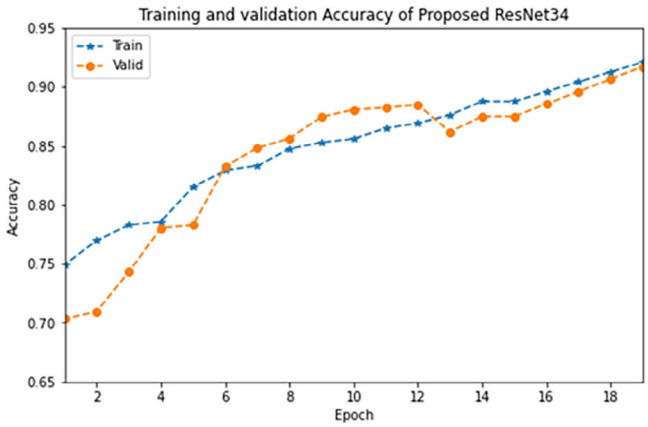

FIGURE 8 Training and validation accuracy graph of the proposed method.

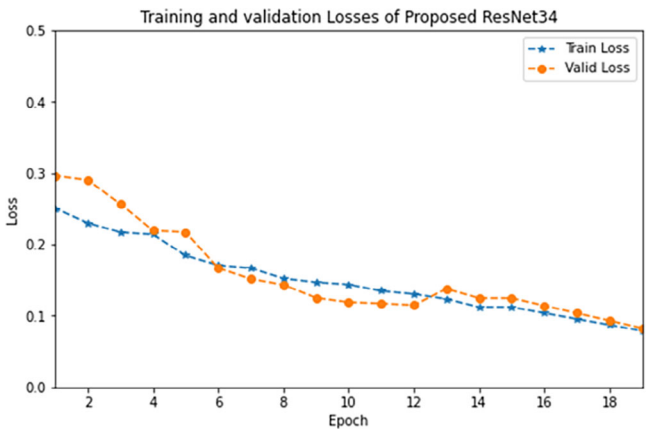

FIGURE 9 Training and validation loss graph of the proposed method.

from the fifth to the sixth epoch but increased drastically after the seventh one, remaining consistent until the twelfth one. The training accuracy continued to improve gradually and reached 91% after the final training. However, the validation accuracy from the seventh to thirteenth epoch was higher than the training accuracy. This was not a result of underfitting but rather a labelling error and the inclusion of training images in the testing data. Utilizing the cleaner widget of FastAi, the images were resolved before the start of the third training phase. At the start of the third phase of training, the validation accuracy dropped slightly but subsequently reached its optimal settings, demonstrating the absence of underfitting or overfitting. At the final epoch, the training and validation accuracy was at the same point, indicating that the model converged sufficiently to detect ASD effectively using facial images in early childhood.

The training and validation losses are illustrated in Figure 9. Upon analyzing the graph, it is evident that the training loss began at 25% and reached 22% at the 3rd epoch. However, at the 5th epoch, the error increased in contrast to the training accuracy, which is inversely proportional. The increase in training accuracy would result in a decrease in error and vice versa. The validation loss began just above 30% and decreased to 22% at the 5th epoch. Subsequently, the training error gradually decreased and did not increase until the final epoch, at which point the error was minimized to below 9%. As observed in the validation accuracy, the validation losses were also lower than the training losses due to the same error. Once the error was settled, the validation loss reached its optimal point, and at the end of the training, both losses were at the same point.

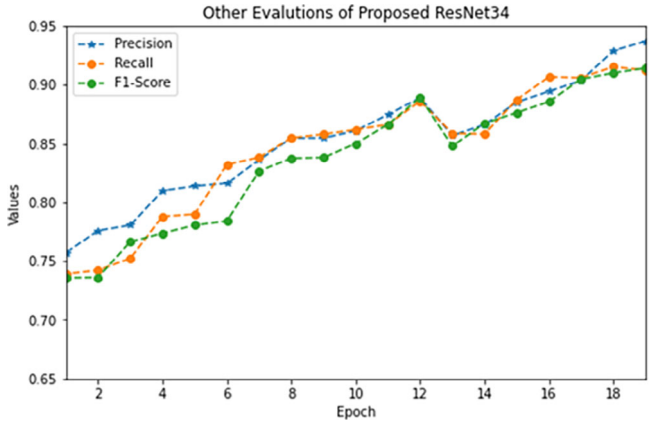

FIGURE 10 Precision, recall, and F1-score graph of the proposed method.

Precision, recall, F1-score, and accuracy scores are also measured to evaluate the performance of the proposed ASDDFI model. Using precision, recall, F1-score and accuracy together provide a more comprehensive evaluation of a model's performance. For example, a high precision score in combination with a low recall score may indicate that the model is conservative in its predictions, while a high recall score in combination with a low precision score may indicate that the model is overpredicting positive instances. The evaluation metrics, accuracy, precision, F1-score, and recall scores of ASD-DFI are depicted in Figure 10. The precision score began at just above 75% and surpassed 88% at the 13th epoch, ultimately reaching 93% after the final epoch. The recall and F1-score began at a point below 75%, but after the 13th epoch, they reached 88%, similar to the precision score. After the final epoch, the recall and F1-score ended at the same point as they began, although variations were observed during the intermediate epochs. Both metrics ended at 91% after the final epoch (Figure 10). From the analysis of this graph, it is evident that our proposed ASD-DFI model does not exhibit a significant difference in evaluation metric scores, indicating that there is no conservative or overpredicting behaviour observed in the detection of ASD from facial images.

In deep learning models, a confusion matrix is a powerful tool for evaluating the performance of deep learning models, particularly for classification tasks. It can help to identify patterns in the model's predictions, such as which classes are being misclassified and to what degree. This can be useful in identifying areas where the model needs to be improved and can also be used to compare the performance of different models. Similarly, a confusion matrix was utilized to visually estimate a model's performance on testing, as depicted in Figure 11. In the presented confusion matrix, right predictions are exhibited diagonally in green colour, and incorrect predictions off-diagonally in grey

20533713, 2024, 4, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/hit2.12073 by PEC University Of Technology, Wiley Online Library on [23/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 5 Performance comparison of the proposed ASD-DFI with state-of-the-art.

Reference, year	Method	Dataset	Accuracy (%)
[38], 2015	PCA	JAFFE	82.3
[35], 2018	DCNN	FER2013	62.11
[51], 2022	Xception	Autism_Image_Data	91
[52], 2022	Hybrid Naïve bayes	Self-Collected	87.50
[53], 2023	Texture features + SVM	Autism_Image_Data	77.96
[28], 2023	AlexNet	ASD	87
Proposed method	**ResNet50**	**ASD**	**92**

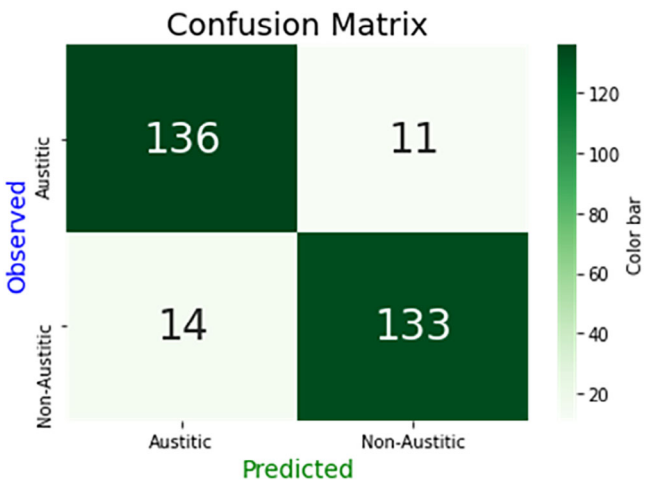

FIGURE 11 Confusion matrix of the proposed method.

colour. The actual labels are presented on *X*-axis in blue colour, and predicted labels are presented on *Y*-axis in green colour.

The confusion matrix illustrates that the ASD-DFI model accurately predicted 136 images belonging to the autistic class, and these are represented in green. This indicates that the model's predictions were consistent with the ground truth for the autistic class on 136 occasions. However, the ASD-DFI model incorrectly classified 11 images as not belonging to the autistic class when they did. This is referred to as misclassification and is classified as true negative. In regards to the non-autistic class, the model accurately predicted 133 samples, which can be classified as false negative. Additionally, only 14 images were incorrectly classified, referred to as false positive (Figure 11). The average accuracy achieved by the ASD-DFI model was 92%. The average accuracy achieved by the ASD-DFI model was 91%. The scores for other evaluation metrics were also measured; the ASD-DFI produced 93% precision, 91% recall, and 91% F1-score. All the evaluation metrics depicted the excellent performance of the proposed method.

4.3 | Performance comparison of the proposed ASD-DFI with state-of-the-art

Despite the fact that a number of deep learning algorithms have been proposed for the classification of ASD, to the best of our knowledge, only one study has employed a transfer learning approach for the detection of ASD using facial images and utilizing the same dataset. As such, our comparison is limited to this study, as it employed the same dataset and utilized the same transfer learning model. However, to critically evaluate our proposed model's performance, we have also incorporated other deep learning studies with top performances. In order to classify ASDs, the suggested ASD-DFI method is compared in Table 5 to current best practices. Using PCA, the JAFFE dataset was accurately classified 82.3% of the time in 2015 (reference [38]). A more recent study in 2018 [35] used DCNN on the FER2013 dataset and achieved 62.11% accuracy. In 2022, there were significant improvements, such as the use of the Xception architecture on Autism Image Data reaching an astounding 91% accuracy and the proposal of a hybrid naïve Bayes strategy on self-collected data reaching 87.50% accuracy [51]. On autism image data, another study conducted in 2022 [53] used a combination of textural features and SVM to achieve an accuracy rate of 77.96%. Using AlexNet on ASD data, [28] achieved an accuracy of 87% in 2023. Using ResNet50 on ASD data, the suggested approach achieves an impressive 92% accuracy, surpassing the state-of-the-art. In comparing several approaches and datasets for ASD classification, this comparison highlights the effectiveness of the suggested ASD-DFI method. It demonstrates the significance of our proposed model in detecting autism using facial images and its potential for efficient use in detecting ASD in early childhood.

5 | CONCLUSIONS

In the context of this study, the performance of well-established pre-trained models for the task of autism spectrum disorder detection was evaluated. The characteristics of these models, such as the relationship between resolution and performance, the number of CNN layers, the number of epochs, and the learning parameters, were investigated. It is clear that pre-trained models can be adapted for the task of autism disorder detection with a limited number of training samples. With regards to the effect of the number of CNN layers on the accuracy, although the research did not include a search for the optimal number of layers for the task, models with 50 and 34 layers achieved a high accuracy rate compared to MobileNetV2, with more than 53 layers. In conclusion, pre-trained networks

20533713, 2024, 4, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/hit2.12073 by PEC University Of Technology, Wiley Online Library on [23/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

possess high applicability for autism disorder detection, even when they were previously trained on different datasets. It was observed that the features learned during training are transferable to other tasks with a high accuracy rate. Additionally, the reduced number of training samples required and the swift convergence of the models make pre-trained models a viable option for applying CNNs to the task of autism disorder detection. In future we will develop a mobile application based on the proposed model for autism disorder detection.

ACKNOWLEDGEMENTS

The authors are highly grateful to their affiliated universities and institutes for providing research facilities.

CONFLICT OF INTEREST STATEMENT

The authors have no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data will be available upon request to the corresponding author.

ORCID

Javed Rashid https://orcid.org/0000-0003-3416-9720
Muhammad Faheem https://orcid.org/0000-0003-4628-4486

REFERENCES

1. Newschaffer, C.J., Croen, L.A., Daniels, J., et al.: The epidemiology of autism spectrum disorders. Annu. Rev. Public Health 28, 235 (2007)
2. Miles, J.H.: Autism spectrum disorders: a genetics review. Genet. Med. 13(4), 278–294 (2011)
3. Saresella, M., Piancone, F., Marventano, I., et al.: Multiple inflammasome complexes are activated in autistic spectrum disorders. Brain Behav. Immun. 57, 125–133 (2016)
4. Erkan, U., DN, T.: Autism spectrum disorder detection with machine learning methods. Curr. Psychiatry Res. Rev. 15(4), 297–308 (2019)
5. Stevens, E., Dixon, D.R., Novack, M.N. et al.: Identification and analysis of behavioral phenotypes in autism spectrum disorder via unsupervised machine learning. Int. J. Med. Inf. 129, 29–36 (2019)
6. Maenner, M.J., Shaw, K.A., Bakian, A.V. et al.: Prevalence and characteristics of autism spectrum disorder among children aged 8 years autism and developmental disabilities monitoring network, 11 sites, United States. MMWR Surveill. Summ. 70(11), 1–14 (2018)
7. Zaroff, C.M., Uhm, S.Y.: Prevalence of autism spectrum disorders and influence of country of measurement and ethnicity. Soc. Psychiatry Psychiatr. Epidemiol. 47(3), 395–398 (2012)
8. Zablotsky, B., Black, L.I., Maenner, M.J. et al.: Prevalence and trends of developmental disabilities among children in the united states 2009–2017. Pediatrics 144(4), e20190811 (2019)
9. Abrahams, B.S., Geschwind, D.H.: Advances in autism genetics: on the threshold of a new neurobiology. Nat. Rev. Genet. 9(5), 341–355 (2008)
10. Betancur, C.: Etiological heterogeneity in autism spectrum disorders: more than 100 genetic and genomic disorders and still counting. Brain Res. 1380, 42–77 (2011)
11. Raj, S., Masood, S.: Analysis and detection of autism spectrum disorder using machine learning techniques. Proc. Comput. Sci. 167, 994–1004 (2020)
12. Zauderer, S.: 69+ autism statistics: how many people have autism? https://www.crossrivertherapy.com/autism-statistics (2022). Accessed 2 May 2022
13. Loftus, Y.: Autism statistics you need to know in 2022. https://www.autismparentingmagazine.com/autism-statistics/ (2022). Accessed 29 Dec 2023
14. Goh, K.L., Morris, S., Rosalie, S., et al.: Typically developed adults and adults with autism spectrum disorder classification using centre of pressure measurements. In: *2016 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 844–848. IEEE, Piscataway, NJ (2016)
15. Andy, C., Masters, R.S.: Improving motor skill acquisition through analogy in children with autism spectrum disorders. Psychol. Sport Exercise 41, 63–69 (2019)
16. Ali, G., et al.: Lyme rashes disease classification using deep feature fusion technique. Skin Res. Technol. 29(11), e13519 (2023)
17. Akram, A., et al.: Segmentation and classification of skin lesions using hybrid deep learning method in the Internet of Medical Things. Skin Res. Technol. 29(11), e13524 (2023)
18. Sohail, M., et al.: Racial identity-aware facial expression recognition using deep convolutional neural networks. Appl. Sci. 12(1), 88 (2021)
19. Wolf, K.: Measuring facial expression of emotion. Dialogues Clin. Neurosci. 17(4), 457–462 (2022)
20. Verma, S.: Sentiment analysis of public services for smart society: literature review and future research directions. Gov. Inf. Q. 39(3), 101708 (2022)
21. Alarood, A.A., et al.: Secure medical image transmission using deep neural network in e-health applications. Healthcare Technol. Lett. 10(4), 87–98 (2023)
22. Ullah, KA., et al.: Machine learning-based prediction of osteoporosis in postmenopausal women with clinical examined features: a quantitative clinical study. Health Sci. Rep. 6(10), e1656 (2023)
23. Shervan, F.E.: Gender classification in human face images for smart phone applications based on local texture information and evaluated Kullback-Leibler divergence. Trait. Signal 36(6), 507–514 (2019)
24. Rashid, J., et al.: Mouth and oral disease classification using InceptionResNetV2 method. Multimedia Tools Appl. 1(9), 1–19 (2023). https://doi.org/10.1007/s11042-023-16776-x
25. Ali, G., Dastgir, A., Iqbal, M.W., Anwar, M.: A hybrid convolutional neural network model for automatic diabetic retinopathy classification from fundus images. IEEE J. Transl. Eng. Health Med. 11, 341–350 (2023)
26. Khan, A.A., et al.: D2PAM: Epileptic seizures prediction using adversarial deep dual patch attention mechanism. CAAI Trans. Intell. Technol. 8(3), 755–769 (2023)
27. Setien-Ramos, I., Lugo-Marín, J., Gisbert-Gustemps, L., et al.: Eye-tracking studies in adults with autism spectrum disorder: a systematic review and meta-analysis. J Autism Dev. Disord. 1–14 (2022)
28. Ghazal, T.M., Issa, G.: Alzheimer disease detection empowered with transfer learning. Comput., Mater. Continua 70(3), 5005–5019 (2022)
29. Fernandez-Duque, D., Black, S.E.: Impaired recognition of negative facial emotions in patients with frontotemporal dementia. Neuropsychologia 43(11), 1673 (2005)
30. Sweeney, M.D., Kisler, K., Montagne, A., et al.: The role of brain vasculature in neurodegenerative disorders. Nat. Neurosci. 21(10), 1318 (2018)
31. Yolcu, G., Oztel, I., Kazan, S., et al.: Facial expression recognition for monitoring neurological disorders based on convolutional neural network. Multimedia Tools Appl. 78(22), 31581 (2019)
32. Guha, T., Yang, Z., Ramakrishna, A., et al.: On quantifying facial expression-related atypicality of children with autism spectrum disorder. In: *2015 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 803–807. IEEE, Piscataway, NJ (2015)
33. Grossard, C., Hun, S., Dapogny, A., et al.: Teaching facial expression production in autism: the serious game JEMImE. Creative Educ. 10(11), 2347–2366 (2019)
34. Dapogny, A., Grossard, C., Hun, S., et al.: JEMImE: a serious game to teach children with ASD how to adequately produce facial expressions. In: 2018 13th IEEE International Conference on Automatic Face & Gesture Recognition (FG 2018), pp. 723–730. IEEE, Piscataway, NJ (2018)
35. Haque, M.I.U., Valles, D.: A facial expression recognition approach using DCNN for autistic children to identify emotions. In: 2018 IEEE 9th Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON), pp. 546–551. IEEE, Piscataway, NJ (2018)
36. Di Nuovo, A., Conti, D., Trubia, G., et al.: Deep learning systems for estimating visual attention in robot-assisted therapy of children with autism and intellectual disability. Rob. 7(2), 25 (2018)
37. Singh, S., Ramya, R., Sushma, V., et al.: Facial recognition using machine learning algorithms on Raspberry Pi. In: 2019 4th International

20533713, 2024, 4, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/htl2.12073 by PEC University Of Technology, Wiley Online Library on [23/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

Conference on Electrical, Electronics, Communication, Computer Technologies and Optimization Techniques (ICEECCOT), pp. 197–202. IEEE, Piscataway, NJ (2019)
38. Smitha, K.G., Vinod, A.P.: Facial emotion recognition system for autistic children: a feasible study based on FPGA implementation. Med. Biol. Eng. Comput. 53(11), 1221–1229 (2015)
39. Wu, S.X., Wai, H.T., Li, L., et al.: A review of distributed algorithms for principal component analysis. Proc. IEEE 106(8), 1321–1340 (2018)
40. Pramerdorfer, C., Kampel, M.: Facial expression recognition using convolutional neural networks: state of the art. arXiv:161202903 (2016)
41. Kaly-Ibala, R., Mabiala-Babela, J., Senga, P.: Encephalopathy after pertussis immunization. Arch. Pediatr. 7(2), 216 (2000)
42. Niu, K., Guo, J., Pan, Y., et al.: Multichannel deep attention neural networks for the classification of autism spectrum disorder using neuroimaging and personal characteristic data. Complexity 2020, 1–9 (2020)
43. Thabtah, F.: Autism spectrum disorder screening: machine learning adaptation and DSM-5 fulfillment. In: Proceedings of the 1st International Conference on Medical and Health Informatics. pp. 1–6. ACM, New York (2017)
44. Mythili, M., Shanavas, A.: A study on autism spectrum disorders using classification techniques. Int. J. Soft Comput. Eng. 4(5), 88–91 (2014)
45. Li, B., Sharma, A., Meng, J., et al.: Applying machine learning to identify autistic adults using imitation: an exploratory study. PLoS One 12(8), e0182652 (2017)
46. Autism dataset. https://www.kaggle.com/cihan063/autism-image-data.2022 (2022). Accessed 8 Mar 2022
47. Bisong, E.: Building Machine Learning And Deep Learning Models On Google Cloud Platform: A Comprehensive Guide For Beginners, pp. 59–64. ACM, New York (2019)
48. Nvidia data center. https://www.nvidia.com/en-us/data-center/ (2023). Accessed 14 June 2023
49. Howard, J., Gugger, S.: Fastai: a layered API for deep learning. Information 11(2), 108 (2020)
50. Paszke, A., Gross, S., Massa, F., et al.: Pytorch: an imperative style, high-performance deep learning library. In: NIPS'19: Proceedings of the 33rd International Conference on Neural Information Processing Systems, pp. 8026–8037. ACM, New York (2019)
51. Khanna, A., Mishra, M., Pati, U.C.: A hybrid feature based approach of facial images for the detection of autism spectrum disorder. In: International Conference on Data Analytics and Insights, pp. 389–399. Springer, Cham (2023)
52. Liao, M., Duan, H., Wang, G.: Application of machine learning techniques to detect the children with autism spectrum disorder. J. Healthcare Eng. 2022, 1–10 (2022)
53. Alsaade, F.W., Alzahrani, M.S.: Classification and detection of autism spectrum disorder based on deep learning algorithms. Comput. Intell. Neurosci. 2022, 8709145 (2022)

How to cite this article: Ahmad, I., Rashid, J., Faheem, M., Akram, A., Khan, N.A., Amin, R.: Autism spectrum disorder detection using facial images: a performance comparison of pretrained convolutional neural networks. Healthc. Technol. Lett. 11, 227–239 (2024). https://doi.org/10.1049/htl2.12073

20533713, 2024, 4, Downloaded from https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/htl2.12073 by PEC University Of Technology, Wiley Online Library on [23/03/2026]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License